

Programming Assignment - 3

Complete Diagnostic Bug Analysis Report (Problems 1-10)

Problem 1 (problem1.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
1	8	Missing semicolon after declaration of x.	Syntax Error	Change to 'int x = 10;'
2	13	Assignment operator '=' used instead of equality comparison '==' inside if condition.	Logical Error	Change to 'if(x == 10)'
3	18	String literal "Ninety" assigned to integer variable 'marks'.	Type Mismatch Error	Change to 'int marks = 90;'
4	20	Variable 'z' used in printf without being declared.	Semantic Error	Declare 'int z;' before use.
5	22	L-value required: cannot assign value of variable 'x' to a numeric literal constant '100'.	Syntax Error	Change to 'x = 100;'
6	26	Array index 10 exceeds the bounds of array 'arr[5]'.	Array Out of Bounds Error	Use index 0-4 (e.g., 'arr[4] = 50;')
7	28	Address of an integer variable '&x' assigned to a float pointer 'float *ptr'.	Type Mismatch Error	Change to 'int *ptr = &x;'
8	30	Format specifier '%f' used to print integer variable 'x'.	Format Specifier Error	Change to '%d' inside printf.
9	35	Division by zero ('a / b' where 'b = 0') causes a runtime crash.	Runtime Error	Ensure divisor 'b' is non-zero before division.
10	37	Multi-character constant 'AB' assigned to character variable 'ch'.	Semantic Error	Use a single character constant like 'A'.
11	39	Both 'if' and 'else' blocks contain identical execution logic and print statements.	Logical Error	Provide appropriate different logic in the else block.
12	44	Useless expression statement 'sqrt(25);' where return value is completely discarded.	Logical Warning	Assign the result: 'double res = sqrt(25);'
13	46	Function 'show()' called without any prior declaration or prototype definition.	Compilation Error	Declare 'void show();' before main.
14	46	Function 'show()' is called but completely missing a body definition.	Linker Error	Provide definition for 'void show() { }'.
15	48	Semicolon immediately after 'while(x < 15)' creates an empty loop body, resulting in an infinite loop.	Logical Error	Remove the trailing semicolon after the while condition.
16	53	'continue' statement placed outside the body of any active loop construct.	Syntax Error	Remove 'continue;' or place it inside a loop.
17	55	'break' statement placed outside any active loop or switch structure.	Syntax Error	Remove 'break;' or place it inside a loop/switch.

S.No.	Line No.	Error Description	Error Type	Error Correction
18	57	Function 'calculate()' called with only one argument, but prototype mandates two parameters.	Function Argument Error	Pass both arguments, e.g., 'calculate(10, 20);'

Problem 2 (problem2.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
19	8	Missing semicolon after declaration of variable 'num2'.	Syntax Error	Change to 'int num2 = 20;'
20	21	Missing closing parenthesis in if statement condition: 'if(choice == 1'.	Syntax Error	Change to 'if(choice == 1)'
21	33	Extra closing parenthesis in else-if statement condition: 'else if(choice == 3))'.	Syntax Error	Change to 'else if(choice == 3)'
22	51	Missing closing parenthesis in for loop header layout: 'for(int i=0; i<5; i++'.	Syntax Error	Change to 'for(int i=0; i<5; i++)'
23	58	Missing closing parenthesis in while loop header condition: 'while(choice > 0'.	Syntax Error	Change to 'while(choice > 0)'
24	68	Missing trailing semicolon in 'while(choice > 5)' condition of a do-while loop.	Syntax Error	Change to 'while(choice > 5);'
25	75	Missing colon after 'case 2' statement inside the switch block.	Syntax Error	Change to 'case 2:'
26	108	Unterminated character constant or string layout mismatch: 'char grade = 'A'.	Syntax Error	Change to 'char grade = 'A';'
27	112	Missing closing square bracket for array declaration: 'int arr[5'.	Syntax Error	Change to 'int arr[5];'
28	139	Missing semicolon at the end of statement: 'printf("End of Program\n")'.	Syntax Error	Change to 'printf("End of Program\n");'

Problem 3 (problem3.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
29	8	String literal "Twenty" assigned to integer variable 'age'.	Type Mismatch Error	Change to 'int age = 20;'
30	14	String literal "A" assigned to character variable 'grade'.	Type Mismatch Error	Change to 'char grade = 'A';'
31	16	Format specifier '%d' used to display float variable 'salary'.	Format Specifier Error	Change specifier to '%f'.
32	18	Format specifier '%f' used to display integer variable 'marks'.	Format Specifier Error	Change specifier to '%d'.
33	22	Variable 'score' used in arithmetic expression but never declared in scope.	Compilation Error	Declare 'int score = 0;' before use.
34	24	Redefinition of variable 'x' in the same local scope.	Compilation Error	

S.No.	Line No.	Error Description	Error Type	Error Correction
				Remove or rename the duplicate declaration 'int x = 20;'
35	27	L-value required: assignment of variable 'x' to numeric literal constant '100'.	Syntax Error	Change to 'x = 100;'
36	31	Format specifier '%s' used to print an integer variable 'num'.	Format Specifier Error	Change specifier to '%d'.
37	35	Format specifier '%d' used to print float variable 'pi'.	Format Specifier Error	Change specifier to '%f'.
38	39	Array index must be an integer, but a float value '2.5' is supplied.	Compilation Error	Use an integer index like 'arr[2] = 100;'
39	41	Floating point constant '65.5' implicitly narrowed to char variable 'ch'.	Warning / Type Mismatch	Use a proper character or cast: 'char ch = 65;'
40	45	Function 'add()' called with 1 argument instead of 2.	Argument Error	Change to 'add(10, 20);'
41	51	Function 'average()' called with 2 arguments instead of 3.	Argument Error	Change to 'average(10, 20, 30);'
42	55	String literal "35" directly assigned to integer variable 'temperature'.	Type Mismatch Error	Change to 'int temperature = 35;'
43	59	String literal "10.5" assigned to float variable 'discount'.	Type Mismatch Error	Change to 'float discount = 10.5f;'
44	65	Array type 'char name[20]' is not assignable using the operator '=' after initialization.	Compilation Error	Use 'strcpy(name, "Robil");' instead.
45	69	Variable 'totalMarks' used but never declared in this program scope.	Compilation Error	Declare 'int totalMarks;' before using it.
46	73	Format specifier '%c' used to print an integer variable 'number'.	Format Specifier Error	Change specifier to '%d'.
47	77	Format specifier '%d' used to print float variable 'rate'.	Format Specifier Error	Change specifier to '%f'.
48	84	Variable 'c' passed to function 'add' but never declared in this scope.	Compilation Error	Declare 'int c = 30;' before use.
49	93	Format specifier '%d' used to print a float expression result inside area calculation.	Format Specifier Error	Change to '%f'.
50	97	Assignment operator '=' used inside condition instead of comparison '=='.	Logical Error	Change to 'if(studentMarks == 100)'
51	102	String literal "Large" assigned to integer variable 'size'.	Type Mismatch Error	Change to 'char *size = "Large";'
52	106	Format specifier '%d' used to display float variable 'percentage'.	Format Specifier Error	Change specifier to '%.2f'.
53	110	String subscript "five" used as an index for array 'arr2'.	Compilation Error	Use an integer subscript like 'arr2[5] = 100;'

S.No.	Line No.	Error Description	Error Type	Error Correction
54	120	Switch case constant must be an integer/character literal, but string literal "2" is used.	Compilation Error	Change to 'case 2:'
55	125	Function 'multiply()' called but never declared or defined in the program.	Compilation Error	Declare and define 'int multiply(int, int);'
56	131	Format specifier '%d' used to print float variable 'areaRect'.	Format Specifier Error	Change specifier to '%f'.
57	148	Variable 'value' used in printf but never declared in scope.	Compilation Error	Declare 'int value;' before use.
58	150	String literal "EMP101" assigned to integer variable 'employeeId'.	Type Mismatch Error	Change to 'char *employeeId = "EMP101";'

Problem 4 (problem4.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
59	41	External variable 'companyProfit' declared with extern but never defined.	Linker Error	Provide global definition: 'float companyProfit = 0.0f;'
60	43	Function 'generateReport()' called but completely missing a definition.	Linker Error	Define 'void generateReport() { }'.
61	45	Function 'displayEmployee()' called but missing definition.	Linker Error	Define 'void displayEmployee() { }'.

Problem 5 (problem5.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
62	11	Division by zero ('a / b' where 'b = 0') causes crash.	Runtime Error	Validate divisor before operating.
63	15	Dereferencing a explicit NULL pointer '*ptr1 = 50;'.	Runtime Error	Allocate valid memory before dereferencing.
64	19	Array index out of bounds: 'arr[10] = 100;' for an array of size 5.	Runtime Memory Error	Use index 0 to 4.

Problem 6 (problem6.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
65	14	Wrong parity check logic: 'num % 2 == 1' incorrectly prints 'Number is Even'.	Logic Error	Change condition to 'num % 2 == 0'
66	24	Inverted grading thresholds criteria: checking 'marks >= 90' yields 'Grade B' instead of 'Grade A'.	Logic Error	Swap the print statements for Grade A and Grade B.

Problem 7 (problem7.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
67	11	Off-by-one array bounds overflow: loop checks 'i <= 5' accessing index 5 of 'marks[5]'.	Out of Bounds Error	Change condition to 'i < 5'.
68	57	Array variable 'city' cannot be assigned directly using string assignment operator '='.	Compilation Error	Use 'strcpy(city, "Delhi");'

Problem 8 (problem8.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
69	16	Function 'add()' called with only 1 argument instead of 2 parameters.	Argument Mismatch	Change call to 'add(num1, num2);'
70	36	'swap()' function passes variables by value instead of using pointers.	Logic Error	Modify signature to use pointers: 'swap(&x, &y);'

Problem 9 (problem9.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
71	18	Assigning value through dereferencing a NULL pointer: '*ptr2 = 100;'.	Null Pointer Dereference	Point to a valid address before setting content.
72	46	Use-after-free bug: accessing '*ptr5' after it has been returned via 'free(ptr5);'.	Dangling Pointer Error	Do not read pointer content after freeing it.

Problem 10 (problem10.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
73	14	Missing semicolon after variable initialization statement 'int choice = 1'.	Syntax Error	Change to 'int choice = 1;'
74	32	Assignment operator '=' utilized inside conditional statement instead of '=='.	Logic Error	Change check to 'if(a == b)'
75	75	Division by zero math exception anomaly error ('x / y' where 'y = 0').	Runtime Error	Verify divisor state metrics parameters.